Do Hidden Markov Regimes Add Out-of-Sample Value?

Separating fat tails from hidden states in volatility forecasting, density calibration, and tail risk, 1999–2026

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Abstract

Hidden Markov models (HMMs) occupy a storied place in quantitative finance: the Baum–Welch algorithm was co-invented by Leonard Baum, Jim Simons' first partner at Renaissance Technologies. We ask whether the machinery pays its way out of sample—and, when it does, *which ingredient* is doing the work: the hidden Markov dynamics, or merely the fat tails of the implied mixture distribution. We evaluate seven models (Gaussian and Student-t HMMs, rule-based volatility regimes, RiskMetrics EWMA, rolling variance, and GARCH(1,1) with Normal and Student-t innovations) on the S&P 500 and NASDAQ (1999–2026) and WTI crude (1986–2026), in an expanding-window walk-forward covering 5,600–8,900 out-of-sample days per asset that include the 2008 crisis, COVID-2020, and 2022. Models are scored on variance point forecasts (QLIKE, with Diebold–Mariano tests and Hansen–Lunde–Nason model confidence sets), full predictive densities (log score, Berkowitz calibration), and tail risk (VaR at three levels with Kupiec/Christoffersen backtests; expected-shortfall calibration ratios). Three findings emerge. *First*, for variance point forecasts GARCH(1,1) dominates and the HMMs trail significantly—regime discreteness costs accuracy that no emission distribution repairs. *Second*, for tail risk the decisive ingredient is the tail, not the hidden state: at a 1% VaR the Student-t HMM and GARCH-t are the two best-calibrated models on every asset and the only two whose coverage is statistically correct on WTI, while every Gaussian model—including the Gaussian HMM's own mixture—over-breaches; expected-shortfall ratios show the same split (fat-tailed models 1.00–1.11, Gaussian models 1.11–1.25). *Third*, crisis windows reverse the picture in an instructive way: in the 2008 collapse the HMMs breached a 1% VaR on 7–8% of S&P days—their volatility ceiling, estimated on pre-crisis history, could not reach 2008 levels—while recursive-variance models adapted within days. Hidden states retain genuine descriptive value (they discover the leverage effect unsupervised) and deliver the best full-density calibration on the commodity; but the apparent tail-risk superiority of regime mixtures in Gaussian-only comparisons is shown to come from their tails, not their Markov dynamics. Volatility-targeting overlays improve equity Sharpe ratios under every forecaster yet no improvement survives a stationary-bootstrap test. Sophistication is not edge; the useful sophistication is a fat-tailed innovation, and it is available for one extra parameter.

1 Introduction

The romance of regime-switching models in finance begins with a biographical fact: before founding Renaissance Technologies, Jim Simons partnered with Leonard Baum, co-inventor of the algo-

*apati077@ucr.edu. Code, data and every number in this paper are reproducible from `https://github.com/jarvisss007/regime-monte-carlo` (`run_full_study_v2.py`); a CI-run test suite re-verifies the implementations—including recovery of known hidden states, known fat-tailed states, and known GARCH dynamics from synthetic data—on every commit.

rithm (Baum et al., 1970) that made hidden Markov models estimable. The intellectual bet—that markets, like speech, are driven by unobservable states inferable from noisy observations—has attracted researchers ever since (Hamilton, 1989; Rydén et al., 1998; Ang & Bekaert, 2002; Guidolin & Timmermann, 2007; Nystrup et al., 2018).

This paper asks the question an allocator (or a sceptical PhD committee) would ask—*does the machinery pay its way out of sample?*—and pushes it one step further than the usual horse race. A hidden-Markov mixture differs from a one-line variance recursion in two separable ways: its *dynamics* (a discrete persistent chain rather than a smooth filter) and its *distribution* (a mixture, hence fat-tailed, rather than a Gaussian). Comparing a Gaussian HMM only against Gaussian filters confounds the two. We therefore field a 2×2-style design: Gaussian and Student-t HMMs against GARCH(1,1) with Normal and Student-t innovations, alongside three simpler baselines, so that every claim about "regimes" can be checked against a matched-tail alternative.

We also insist that "forecasting volatility" is not one task. A model may forecast the *level* of tomorrow's variance poorly while describing the *shape* of tomorrow's distribution well; for a risk manager the shape—especially the left tail—is the object that matters. Every model is scored on three progressively richer objects: variance point forecasts under QLIKE loss (Patton, 2011), one-day-ahead densities under the log score with Berkowitz calibration tests (Berkowitz, 2001), and Value-at- Risk at three levels with coverage and independence backtests (Kupiec, 1995; Christoffersen, 1998), supplemented by expected-shortfall calibration ratios. Point- and density-forecast comparisons are disciplined by Diebold–Mariano tests (Diebold & Mariano, 1995) and Hansen–Lunde–Nason model confidence sets (Hansen et al., 2011).

Our contribution is an honest accounting in the spirit of Hansen & Lunde (2005), extended to a decomposition: when regime models look good, is it the regimes or the tails? The answer, on twenty-seven years of equity data and forty of commodity data, is: the tails.

2 Models

2.1 Hidden Markov models

Daily returns r_t are modelled as draws from one of $K = 3$ emission distributions selected by a latent first-order Markov chain with transition matrix A. The **Gaussian HMM** uses $\mathcal{N}(\mu_k, \sigma_k^2)$ emissions, estimated by Baum–Welch (EM over scaled forward–backward recursions) implemented from scratch; states are decoded by Viterbi and relabelled by ascending risk. The **Student-t HMM** replaces each emission with a location-scale t: $r_t \sim \mu_k + \sigma_k\, t(\nu_k)$, estimated by the classical normal/inverse-gamma EM augmentation (per-observation tail weights $u_{tk} = (\nu_k + 1)/(\nu_k + z_{tk}^2)$), with ν_k updated by profile likelihood. Fat tails *inside* each regime let the model separate "the regime got riskier" from "a single outlier landed".

Both implementations are validated on synthetic data with known parameters before touching market data: the Gaussian version recovers a known hidden path with $> 85\%$ Viterbi accuracy; the t version correctly identifies which synthetic state is fat-tailed ($\hat{\nu}$ ordering) and reproduces its own quantiles with correct empirical coverage.

For out-of-sample use we maintain filtered probabilities $p_t = P(s_t \mid r_{1:t})$ recursively; with predictive weights $q = p_t A$ the one-day density is the mixture $\sum_k q_k f_k$, its variance follows from mixture moments, its VaR by bisection on the mixture CDF, and its expected shortfall in closed form from component partial expectations.

Table 1: Three-state Gaussian HMM, S&P 500 daily returns, 1999–2026.

state	μ_k (bp/day)	σ_k (ann.)	self-persistence	E[duration]
calm	+11.0	8.8%	0.974	39.1 days
normal	−0.9	18.7%	0.967	30.4 days
turbulent	−15.2	45.6%	0.959	24.5 days

2.2 Comparators

Rule-based regimes: calm/normal/turbulent by terciles of trailing 21-day realised volatility (edges from training data only); forecast = the current regime's training-pool variance. The observable-state analogue of the HMM. **EWMA**: $\hat{v}_{t+1} = 0.94\hat{v}_t + 0.06r_t^2$ (RiskMetrics, 1996). **Rolling**: 21-day trailing variance. **GARCH(1,1)-Normal** and **GARCH(1,1)-t** (Bollerslev, 1986): $\hat{v}_{t+1} = \omega + \alpha r_t^2 + \beta\hat{v}_t$ with Normal or standardised-$t(\nu)$ innovations, estimated by maximum likelihood at each refit; between refits the recursion runs with fixed parameters. Density, VaR and ES for the t variant use the scaled-t closed forms. For Gaussian-density models the log score is an affine transform of QLIKE, so their density ranking must equal their QLIKE ranking; only the mixture and t models can genuinely differ across scoring rules.

3 Data and protocol

Data are daily closes: S&P 500 and NASDAQ composite, January 1999–June 2026 (6,913 returns each; Yahoo Finance), and WTI crude spot, January 1986–June 2026 (10,189 returns; FRED `DCOILWTICO`); the exact series used are cached in the repository. The first 1,260 days form the initial window; all seven models are re-estimated every 63 trading days on an expanding window; filters and recursions update daily; every forecast of day $t+1$ uses only information through day t. This yields 5,653 out-of-sample days per equity index (2004–2026: covering 2008, COVID-2020, and 2022) and 8,929 for WTI (1991–2026). Statistical machinery: QLIKE $L_t = \log\hat{v}_t + r_t^2/\hat{v}_t$; Diebold–Mariano with Bartlett/HAC variance (5 lags); model confidence sets at the 10% level by stationary block bootstrap (Hansen et al., 2011; Politis & Romano, 1994); Kupiec proportion-of-failures and Christoffersen independence tests at VaR levels $\{1\%, 2.5\%, 5\%\}$; ES calibration as the ratio of mean realised loss beyond VaR to mean predicted ES on breach days (1 = calibrated); Berkowitz LR test on probability integral transforms.

4 Results

4.1 What the hidden states discover

Fitted to the full 1999–2026 S&P sample, the Gaussian HMM's states remain economically interpretable rather than mere volatility buckets (Table 1): the calm state combines a +11.0 bp daily drift with 8.8% annualised volatility; the turbulent state −15.2 bp with 45.6%. The leverage effect is discovered unsupervised, consistent with Rydén et al. (1998). The chain still never jumps directly between calm and turbulent. The t-HMM adds a sharper observation: conditional on the regime, returns remain fat-tailed—$\hat{\nu} \approx 7$ in the calm state and $\hat{\nu} \approx 6$ in the turbulent state—so regime switching alone does not exhaust the non-Gaussianity of daily returns. This foreshadows the tail-risk results below.

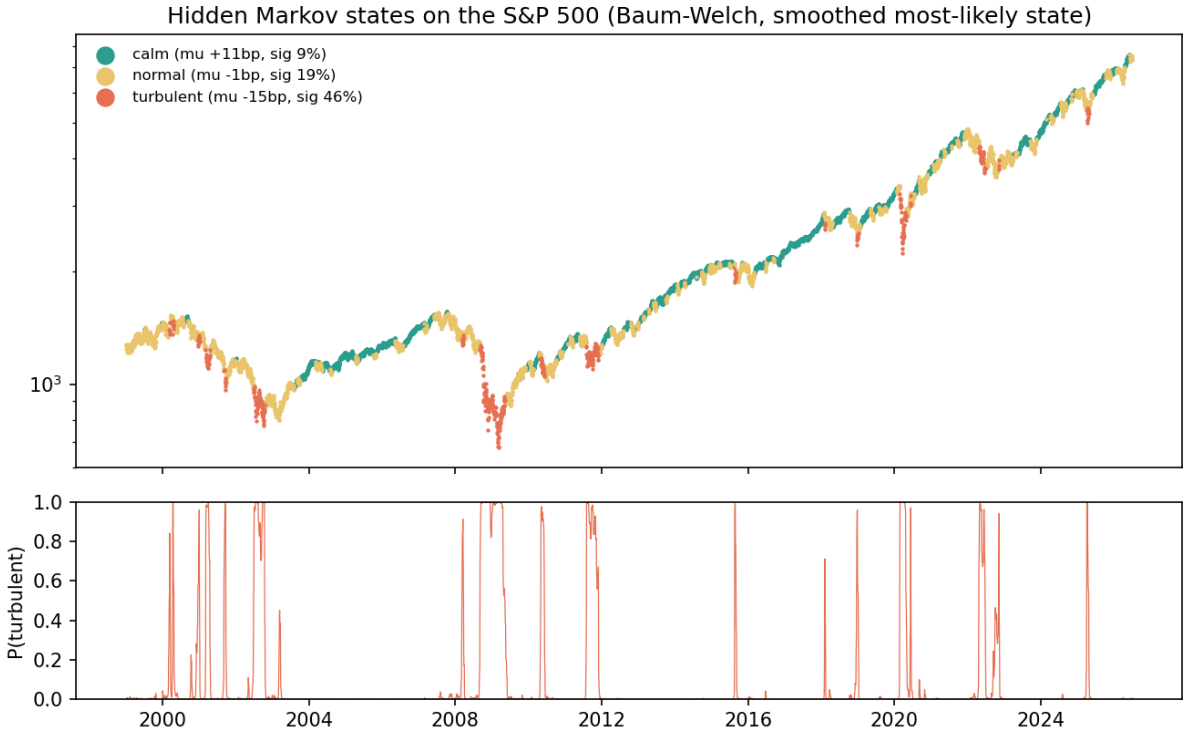


Figure 1: Top: S&P 500 (log scale), 1999–2026, coloured by decoded hidden state. Bottom: smoothed probability of the turbulent state.

4.2 Variance point forecasts: recursions win, tails don't help

Table 2 reports QLIKE. GARCH is best on all three assets, and the innovation distribution is nearly irrelevant to the point forecast (GARCH-N vs GARCH-t: indistinguishable on WTI, GARCH-N marginally ahead on the equities)—as expected, since QLIKE scores only the variance path. Both HMMs trail GARCH-t significantly (DM $t = 2.3$–4.3), and the t-emissions do not repair the gap: discreteness, not the emission distribution, is the binding constraint. The piecewise-constant forecasters (rolling, rule-based) are decisively worst. Model confidence sets tell a sample-dependent story: on NASDAQ the MCS retains *only* GARCH-N; on the S&P and WTI—where the out-of-sample span includes 2008 and COVID volatility of unprecedented scale—the data cannot separate the top six, and only the rule-based forecaster is eliminated.

4.3 Densities: GARCH-t first, t-HMM second

On the log score (Table 3) the fat-tailed pair pulls away from everything Gaussian. GARCH-t is the sole member of the model confidence set on the S&P and WTI and shares a two-member set with the t-HMM on NASDAQ. The t-HMM is second everywhere—ahead of every Gaussian model including GARCH-N—and the Gaussian HMM beats the Gaussian filters on WTI only. Berkowitz calibration tests sharpen the split: on WTI the two HMMs deliver the best-calibrated densities in the study ($p = 0.87$ Gaussian, $p = 0.95$ t; GARCH-N $p = 0.48$, GARCH-t $p = 0.16$; all simple filters rejected at $p < 10^{-10}$), whereas on the equity indices—whose out-of-sample span includes two once-a-generation crises—*every* model's density is rejected. Nothing in this comparison set

Table 2: Variance point forecasts: QLIKE (lower is better); DM t against GARCH-t (positive = GARCH-t more accurate); $\star$ = in the 10% MCS.

	S&P 500		NASDAQ		WTI	
forecaster	QLIKE	t	QLIKE	t	QLIKE	t
GARCH-N	$\mathbf{-8.394}^\star$	-2.1	$\mathbf{-7.966}^\star$	-2.9	$-6.659^\star$	0.1
GARCH-t	$-8.390^\star$	—	-7.963	—	$\mathbf{-6.659}^\star$	—
EWMA(0.94)	$-8.357^\star$	3.9	-7.942	3.8	$-6.639^\star$	3.9
HMM-Normal	$-8.313^\star$	3.3	-7.928	4.3	$-6.612^\star$	2.6
HMM-t	$-8.301^\star$	3.3	-7.944	2.6	$-6.611^\star$	2.3
21d rolling	$-8.301^\star$	5.4	-7.896	5.2	$-6.603^\star$	6.2
rule-based	-8.253	5.1	-7.880	6.8	-6.439	4.6

Table 3: Predictive density: mean log score (higher is better); DM t against GARCH-t on negative log score; $\star$ = in the 10% MCS. Bottom row: Berkowitz calibration p-value on WTI (equity densities are rejected for all models).

	S&P 500		NASDAQ		WTI	
forecaster	score	t	score	t	score	t
GARCH-t	$\mathbf{3.301}^\star$	—	$\mathbf{3.077}^\star$	—	$\mathbf{2.440}^\star$	—
HMM-t	3.281	3.4	$3.073^\star$	1.4	2.431	3.4
GARCH-N	3.278	5.4	3.064	5.3	2.410	5.9
HMM-Normal	3.260	3.6	3.059	4.9	2.416	2.9
EWMA(0.94)	3.260	5.1	3.052	5.5	2.401	5.9
21d rolling	3.231	5.9	3.029	5.9	2.383	7.2
rule-based	3.207	6.8	3.021	8.6	2.301	5.5
Berkowitz p (WTI)					HMM-t .95, HMM-N .87, G-N .48, G-t .16, rest $< 10^{-10}$	

fully describes equity returns through 2008 and 2020; the fat-tailed models merely fail least.

4.4 Tail risk: fat tails, not hidden states

Table 4 is the paper's central evidence. At the 1% level the two Student-t models are the best calibrated on every asset, and on WTI they are the only two whose coverage passes the Kupiec test (GARCH-t hit rate 1.00%, $p = 0.98$; HMM-t 1.09%, $p = 0.42$). Every Gaussian-density model over-breaches everywhere—*including the Gaussian HMM*, whose mixture provides some tail mass but not enough. On the equity indices, whose evaluation span contains 2008 and 2020, no model attains correct 1% coverage, but the ordering is unchanged: t-tails first (hit rates 1.4%–1.7%), Gaussian mixtures next, Gaussian recursions last (up to 2.7%—nearly triple the promised breach rate). At the 5% level, where tail shape matters less, most models pass and the differences fade, confirming that the 1% results are about distribution tails rather than variance levels. Expected-shortfall calibration tells the same story with severity instead of frequency: on breach days the fat-tailed pair's ES ratios sit at 1.00–1.11 across assets while the Gaussian models' realised losses run 11–25% beyond their predicted ES.

The decomposition is now clean. In a Gaussian-only comparison (as in an earlier version of

Table 4: VaR backtests: empirical breach rates (%) at nominal 1% and 5% levels. Bold = Kupiec coverage not rejected at 5% significance. OOS days: 5,653 (equities), 8,929 (WTI).

	S&P 500		NASDAQ		WTI	
forecaster	1%	5%	1%	5%	1%	5%
HMM-t	1.40	**5.41**	1.38	**5.55**	**1.09**	**5.25**
GARCH-t	1.47	**5.27**	1.68	**5.43**	**1.00**	**5.23**
HMM-Normal	1.59	**5.38**	1.82	**5.73**	1.22	**5.38**
GARCH-N	1.86	**5.11**	1.86	**5.25**	1.51	**4.64**
EWMA(0.94)	2.28	5.80	2.07	5.93	1.85	**5.39**
21d rolling	2.65	6.30	2.42	6.51	2.09	5.85
rule-based	2.62	**5.41**	1.98	**4.97**	2.27	**5.48**

Table 5: 1% VaR breach rates (%) inside stress windows, S&P 500 and WTI. GFC: Jul 2008–Jun 2009 (252 days). COVID: Feb–Jul 2020 (≈126 days). 2022: calendar year.

		S&P 500		WTI
forecaster	GFC	COVID	2022	COVID
HMM-Normal	7.9	4.0	1.2	6.4
HMM-t	6.7	4.0	0.8	5.6
rule-based	8.3	7.9	1.6	12.0
EWMA(0.94)	2.0	5.6	1.6	4.8
21d rolling	2.0	5.6	2.0	4.0
GARCH-N	2.4	4.0	1.2	3.2
GARCH-t	1.2	3.2	0.8	3.2

this study), the Gaussian HMM appears to be the best tail model available and one might credit the hidden states. Field a matched-tail benchmark and the credit moves: **one extra degree of freedom—ν—in a one-line GARCH recursion buys tail calibration equal to (on WTI, slightly better than) the full Baum–Welch apparatus.** The hidden states are not what fixes the tail; the tail is.

4.5 Crisis windows: where the state ceiling binds

Sub-sampling the 1%-VaR breaches to three stress windows exposes the HMMs' characteristic failure mode (Table 5). In the second half of 2008 on the S&P, the Gaussian and t HMMs breached on 7.9% and 6.7% of days respectively—seven times their nominal rate—while GARCH-t breached on 1.2% and even the humble EWMA on 2.0%. The mechanism is structural: an HMM's worst-case one-day distribution is bounded by its most turbulent state, whose σ_k was estimated on *pre-2008* history; when realised volatility blew through every level in the training sample, the discrete state ceiling bound, whereas recursive-variance models scale without limit and adapted within days. By COVID-2020 the HMMs had a 2008 in their training window and their breach rates (4.0%) fell in line with the pack; in 2022 every model was calibrated. The same pattern appears in the WTI COVID window (which contains the negative-price episode): recursions 3.2%, HMMs 5.6–6.4%. Regime models learn new regimes only after living through them.

4.6 State-count sensitivity

Re-running the S&P walk-forward with $K \in \{2, 3, 4, 5, 6, 8\}$ Gaussian states (refit every 126 days) improves both QLIKE and log score monotonically in K, with the gains plateauing beyond $K \approx 5$: QLIKE moves from -8.249 (K=2) through -8.344 (K=4) to -8.383 (K=8), closing roughly nine-tenths of the distance to the GARCH benchmark (-8.394) but never crossing it. This is consistent with the discretisation interpretation: a finer state grid approximates a continuous volatility process ever better, converging toward—without reaching—the smooth-recursion frontier. The three-state model's point-forecast deficit is a resolution cost, not evidence against latent structure per se; but no resolution within reach makes the latent chain a *better* point forecaster than the recursion it approximates.

4.7 Volatility targeting, honestly tested

Finally, a 10%-vol-target overlay ($w_t = \min(1, \sigma_{\text{tgt}}/\hat{\sigma}_t)$, 2 bps per side on turnover; Moreira & Muir, 2017) improves the Sharpe ratio over buy-and-hold on both equity indices under *every* forecaster (S&P: 0.54 → 0.58–0.67; NASDAQ: 0.63 → 0.66–0.76) and cuts maximum drawdowns roughly in half (S&P $-57\% \to -25\%$; NASDAQ $-56\% \to -20\%$); on WTI (buy-and-hold drawdown -94%) the overlays reduce drawdown by 50+ points but lower the Sharpe. A stationary-bootstrap test of each Sharpe difference rejects nothing (minimum $p = 0.09$, rule-based on NASDAQ). The drawdown reduction is mechanical and large; the Sharpe improvement remains statistically unproven at this sample size—and offers no wedge for sophisticated forecasters over simple ones.

5 Discussion

The design separates two ingredients that regime-switching folklore bundles together, and they land on opposite sides of the ledger. *The tail is first-order.* Moving from Normal to Student-t innovations—one parameter—repairs VaR calibration for GARCH and sharpens it for the HMM; no Gaussian model of any structural sophistication achieves what either t model achieves. *The hidden state is second-order for prediction.* It costs accuracy at the point forecast (discreteness), adds little to tail calibration beyond what its mixture tails already provide, and imposes a crisis-specific fragility: state ceilings estimated on tranquil history cannot span an unprecedented shock, precisely when a risk model earns its keep. What the hidden state buys instead is *understanding*: unsupervised discovery of the leverage effect, persistent interpretable regimes, the no-direct-jump topology of market stress, and—on the commodity—the best-calibrated full density in the study. For a risk desk, the practical ranking is GARCH-t first for production, t-HMM close behind if regime interpretation is wanted; for a researcher, the lesson is that any claimed benefit of regime dynamics should be benchmarked against a matched-tail alternative before the dynamics get the credit. The Renaissance lesson survives in amended form: Baum's mathematics genuinely sees structure that filters do not—but seeing structure and forecasting from it are different trades.

6 Limitations

(i) The crisis-window analysis is descriptive (three windows, small breach counts). (ii) $K \leq 8$ and hard state assignments at refit boundaries; regime-switching GARCH (persistent states *with* within-state recursions) is the natural hybrid our results motivate, and the obvious next model to field. (iii) Quarterly refits are a compute compromise. (iv) One-day horizon and long-only

overlays; multi-day distributions and derivatives applications may weight the HMM's full-path simulation abilities more heavily. (v) Yahoo/FRED daily closes; results at intraday frequencies may differ.

7 Conclusion

Evaluated honestly, out of sample, across assets and across three scoring rules, hidden Markov models on daily returns are: significantly worse point forecasters than GARCH; competitive density forecasters when given t emissions (and the best-calibrated densities on the commodity); excellent tail-risk models—but no better than a GARCH-t whose single extra parameter supplies the same fat tail without the state ceiling that failed in 2008. The regime machinery's durable contribution is descriptive. The durable practical lesson is cheaper than the machinery: model the tail, and let the variance be recursive.

Reproducibility

`python run_full_study_v2.py` reproduces every table (per-asset checkpoints in `results/v2_*.csv`); `run_ksweep_v2.py` reproduces Section 4.6; `plots_hmm.py` reproduces Figure 1. The cached input data ship in `data/`. `pytest` (20 tests) verifies the HMM, t-HMM, GARCH, VaR/ES and calibration-test implementations against synthetic data with known properties, in CI on every commit.

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